

Assignment 1: Temporal Modeling

Task 1: Heterogeneous Sensors | Task 2: Autoformer Challenge

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AI651 – Deep Learning for Space, Time and Graphs

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GitHub: <https://github.com/msbasit/pa1-report>

Summary. Task 1 shows that explicit context-estimated period routing beats the listed neural forecasters, although decomposition and delay mixing substantially improve neural accuracy. Task 2 applies a small Autoformer encoder–decoder to an anonymized, heavy-tailed series: the optional known-future covariates make the largest measured improvement. The selected three-member ensemble has local validation RMSE 60.173, with 8,403 trainable parameters and 24 production epochs in total. No hidden-target or public leaderboard score is present in the saved evidence.

Evidence and scope. This report summarizes the existing Task 1 report and saved outputs of `Task-2.ipynb` in `q2-colab`; it includes all 15 embedded figures, every reported experiment, and the hypothesis/discussion sequence. Numbers are rounded, not newly estimated by training. Task 1 uses physical units; Task 2 units and the physical sampling interval are undisclosed. Neither notebook was rerun to prepare this report.

1 Task 1 – Forecasting across heterogeneous sensors

1.1 Setup and implementation

Four stations sample vibration at 20 Hz. Products A/B/C operate at periods 14–18/21–27/28–36 samples. Each signal combines smooth drift, vibration, and noise. A model observes one station’s $L = 96$ samples and predicts $H = 48$. Stations 1–3 fit models; station 4 supplies inference context but is excluded from fitting and scaling. Target-time intervals are training $[0, 9600)$, validation $[9600, 12480)$, and test $[12480, 15360)$; windows never cross product changes. Last-value anchoring and training-only scaling are inverted before scoring.

Saved setup: `full`, data/model seed 0, CPU, 6,000 steps per neural fit. The saved checks pass for the world, Raw Attention assembly/gradients, decomposition, decomposed Attention, FFT delay scores, aggregation, and all four unified configurations. Raw Attention embeds, adds positions, applies two blocks, normalizes, flattens, and predicts $[B, H]$. Decomposition uses centered moving averages with replicated endpoints; aggregation indexes $(t - \tau) \bmod L$ and preserves gradients through values and weights.

1.2 Response 1A – Hypothesis, comparison, and revision (Output 1.1)

Hypothesis. Period-routed ridge < Raw Attention < shared ridge $\approx$ sensor-specific ridge on established stations; the same order without sensor-specific ridge on station 4. Period-estimated routing explicitly selects the relevant pace-band map; station identity does not resolve changing pace. **Timing disclosure:** the notebook hypotheses were entered after saved results existed; they are retrospective, not preregistered predictions.

Table 1: Task 1 aggregate validation results and fitting cost (Outputs 1.1/4.1). RMSE in m/s^2 ; MSE in $(\text{m/s}^2)^2$. These are not inference-time measurements.

Model	Est. RMSE	Est. MSE	Held RMSE	Held MSE	Parameters	Fit s
Shared ridge	.767	.588	.876	.767	4,608	.002
Sensor-specific ridge	.768	.590	–	–	13,824	.003
Period-routed ridge	.166	.028	.173	.030	59,904	.006
Raw Attention	.431	.186	.612	.374	368,368	760.032
Attention + decomposition	.296	.087	.430	.185	373,024	897.249
Raw delay mixer	.221	.049	.340	.116	368,370	899.230
Autoformer-inspired	.199	.040	.239	.057	373,026	1008.758

Table 2: Task 1 product-wise validation RMSE (Outputs 1.1/4.1); sensor-specific ridge is unavailable for the held-out station.

Model	Established			Held-out		
	A	B	C	A	B	C
Shared ridge	.805	.854	.621	.909	.982	.713
Sensor-specific ridge	.804	.852	.630	—	—	—
Period-routed ridge	.181	.164	.152	.184	.182	.150
Raw Attention	.388	.482	.418	.647	.580	.607
Attention + decomposition	.264	.335	.283	.359	.488	.434
Raw delay mixer	.211	.211	.239	.415	.281	.309
Autoformer-inspired	.226	.188	.182	.270	.221	.224

The aggregate ranking matches the hypothesis. Shared and sensor-specific ridge are effectively tied and swap their marginal order across products. Compared with shared ridge, routed ridge reduces RMSE by approximately 78% established/80% held-out; Attention reduces it by 44%/30%. Context-dependent mixing helps, but explicit period routing is the better bias in this run.

1.3 Response 1B – Context dependence (Outputs 1.2–1.3)

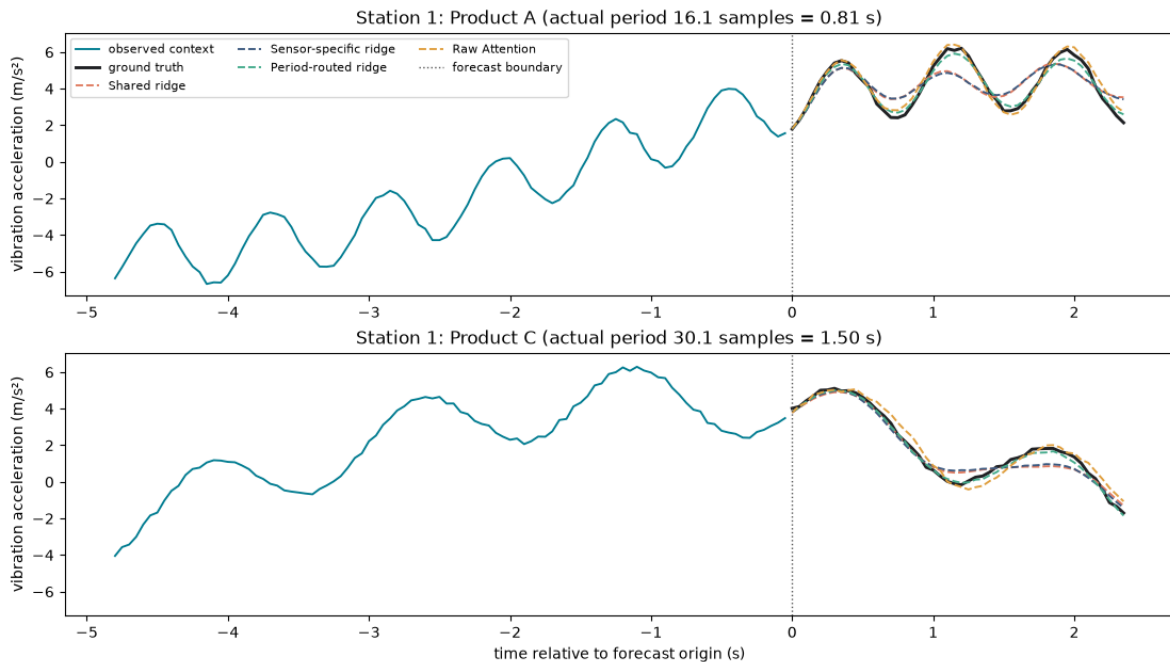


Figure 1: Output 1.2: Station 1 under two different operating paces. All plotted forecasts are validation examples.

Station 1 has periods 16.135 and 30.093 samples under A and C. Fixed ridge attenuates later oscillations/rebounds because one map compromises across incompatible phase advances. Window RMSE for A/C is shared .779/.448, sensor-specific .787/.436, routed .232/.150, and Attention .290/.378. These selected windows are illustrative, not aggregate evidence.

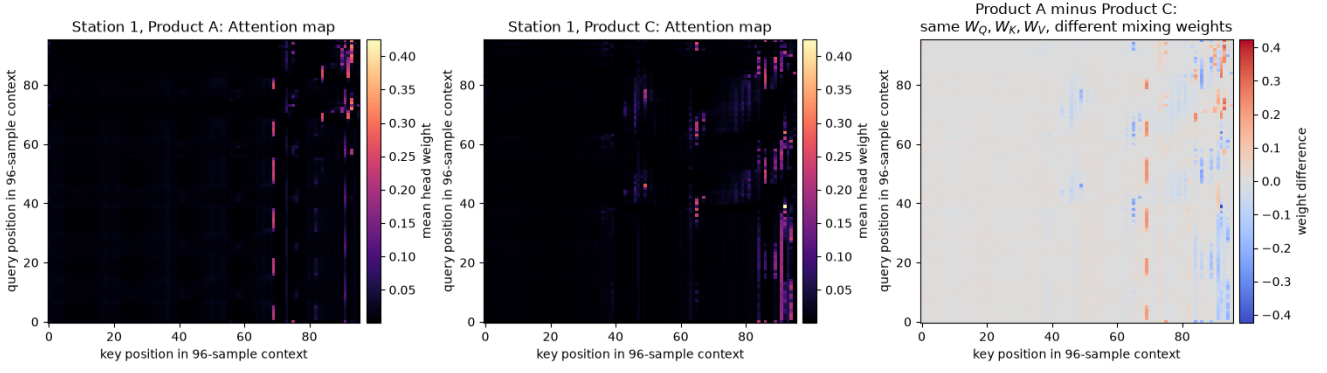


Figure 2: Output 1.3: context-dependent Attention weights. Different mixing weights do not establish recovery of the physical period.

The projections W_Q, W_K, W_V remain fixed, but the current context changes Q, K and hence $\text{softmax}(QK^\top/\sqrt{d_h})$. The mean/max absolute cross-context weight difference is .0113/.4250; a fixed ridge matrix cannot rebuild its mixing rule this way.

1.4 Response 2 – Decomposition (Outputs 2.1–2.3)

$$T_t = \frac{1}{m} \sum_{u=-q}^q x_{t+u}, \quad m = 2q + 1, \quad S = x - T.$$

Endpoints repeat the nearest observed value, giving same-length streams and exact reconstruction up to floating-point tolerance. For $x = [1, 2, 3, 10, 5]$ and $m = 3$, the checked trend is $[1.3333, 2, 5, 6, 6.6667]$, remainder $[-.3333, 0, -2, 4, -1.6667]$. The seasonal encoder and linear trend head specialize; decomposition follows residual sublayers.

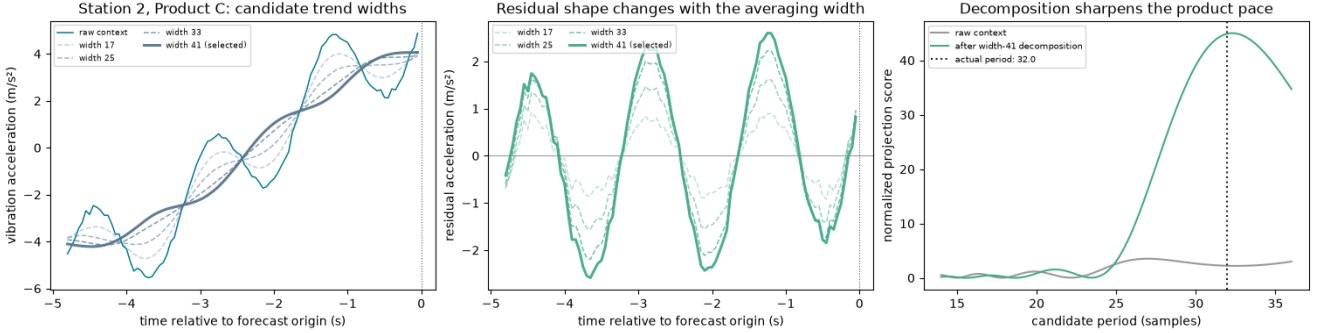


Figure 3: Output 2.1: wider filters follow slower structure and preserve more oscillation in the residual; boundary replication affects endpoint shapes.

Table 3: Outputs 2.1–2.2: filter statistics and established-training oracle recovery.

Representation	Residual std	Trend range	Recovery within 1 sample	Median error
Raw	—	—	68.67%	.525
Width 17	.578	8.768	98.00%	.157
Width 25	1.012	8.037	98.92%	.122
Width 33	1.372	8.076	99.33%	.142
Width 41 (selected)	1.572	8.280	99.85%	.216

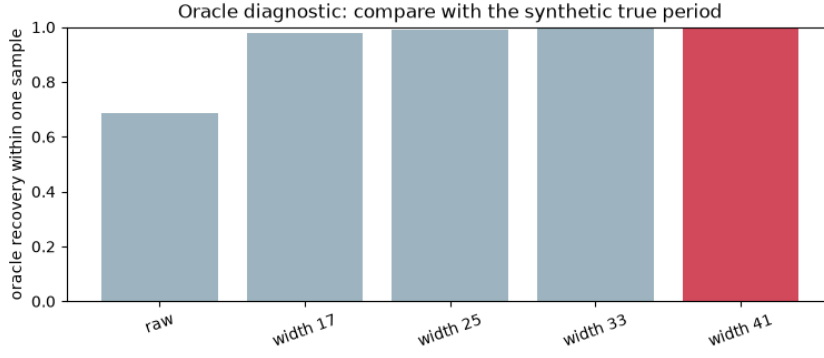


Figure 4: Output 2.2: known synthetic period is used only to score training-window recovery, not as an input to the neural forecaster.

Width 41 (2.05 s) maximizes within-one-sample recovery; width 25 instead minimizes median error. The 240-sample drift scale makes smoothness a sensible bias relative to 14–36-sample vibration. The residual is not the true generative seasonal component. An abrupt level shift would be smeared, leaving transient residuals that a repetition encoder could misread.

Table 4: Output 2.3: decomposition ablation MSE/MAE. Its RMSEs are in the product table above.

Population	Product	Raw Attention		Attention + decomposition	
		MSE	MAE	MSE	MAE
Established	A	.150	.293	.070	.204
Established	B	.233	.358	.112	.249
Established	C	.175	.306	.080	.212
Held-out	A	.418	.502	.129	.275
Held-out	B	.336	.439	.238	.357
Held-out	C	.369	.456	.189	.330

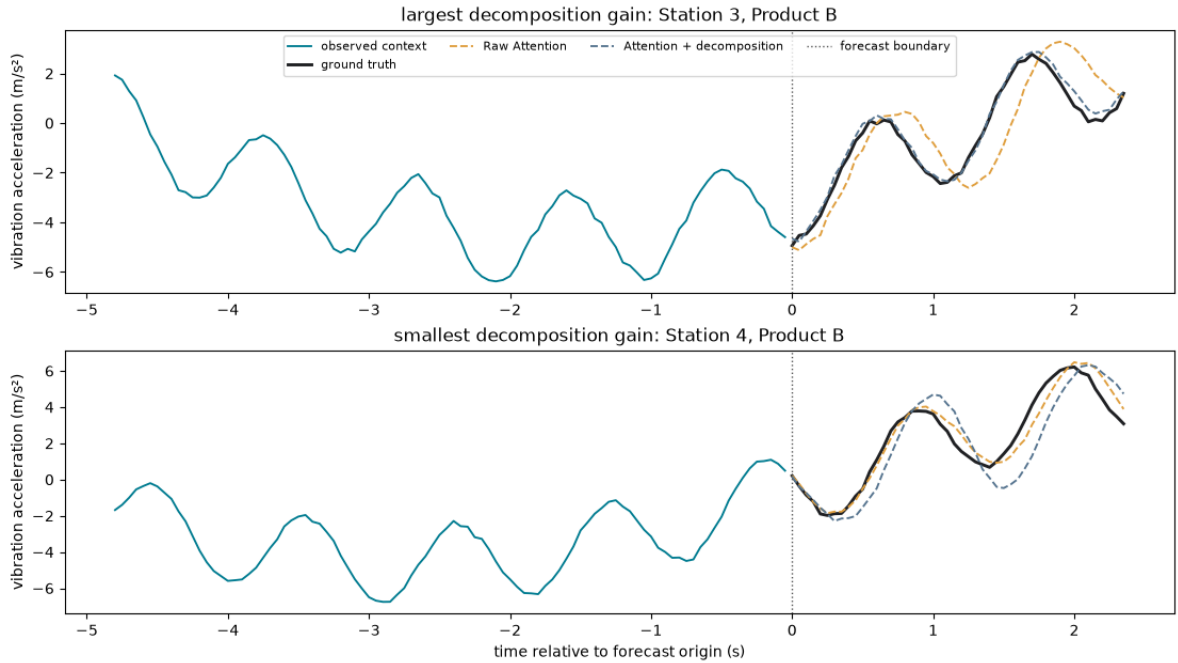


Figure 5: Output 2.3: contrasting decomposition examples; gains are heterogeneous even though each aggregate product/population comparison improves.

Aggregate Attention RMSE falls $.431 \rightarrow .296$ established and $.612 \rightarrow .430$ held-out (approximately 31%/30% reductions).

1.5 Response 3 – Delay mixing (Outputs 3.1–3.2)

$$R(\tau) = \sum_t \tilde{q}_t \tilde{k}_{(t-\tau) \bmod L}, \quad z_t = \sum_{j=1}^K \alpha_j v_{(t-\tau_j) \bmod L}.$$

The supplied Task 1 score centers q, k in time (and averages across heads/features). For $q = k = [1, 3, 1, 3]$, the centered sequence is $[-1, 1, -1, 1]$ and the FFT score matches direct correlation $[4, -4, 4, -4]$ at lags 0–3. The aggregation example $v = [10, 20, 30, 40]$, delays $[1, 3]$, weights $[\text{.75}, \text{.25}]$ returns $[35, 15, 25, 25]$; $z_0 = \text{.75}(40) + \text{.25}(20) = 35$ checks shift direction. Selected delays are per example and shared across heads/features. Task 1 uses the 8–48 lag band, $K = \lceil 2 \ln L \rceil = 10$, and a learned positive softmax temperature.

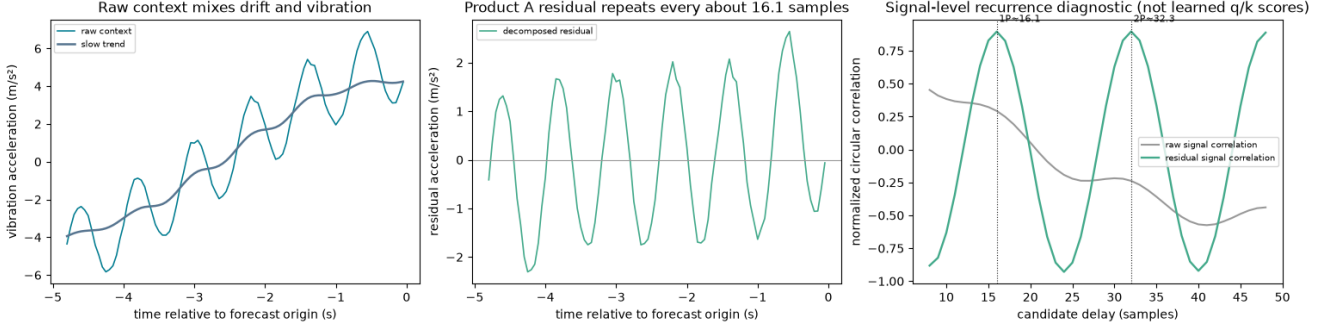


Figure 6: Output 3.2: correlation of raw/residual signal values, not learned query/key scores.

For Station 2/Product A, $P = 16.135$: recurrence near 16 and 32 samples ($P, 2P$) has correlation .896/.897, with a further peak near 48 ($3P$). Removing drift clarifies recurring displacements. An isolated transient or rapidly changing period violates that bias; circular aggregation can spread a unique event into unrelated positions.

1.6 Response 4 – Compare designs and deploy (Outputs 4.1–4.3)

4(a), retrospective hypothesis (two sentences). Decomposition should help most when slow structure is prominent, freeing the encoder from the ramp and exposing repetition. Delay mixing alone still sees a raw representation, so its benefit was expected to be less direct.

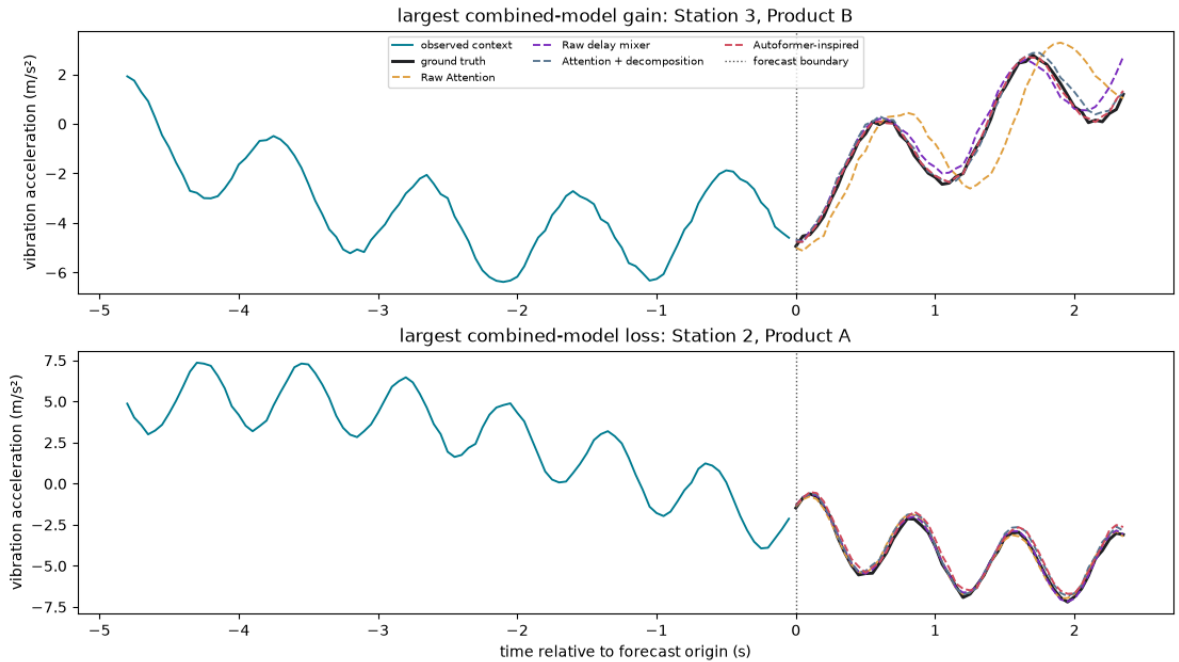
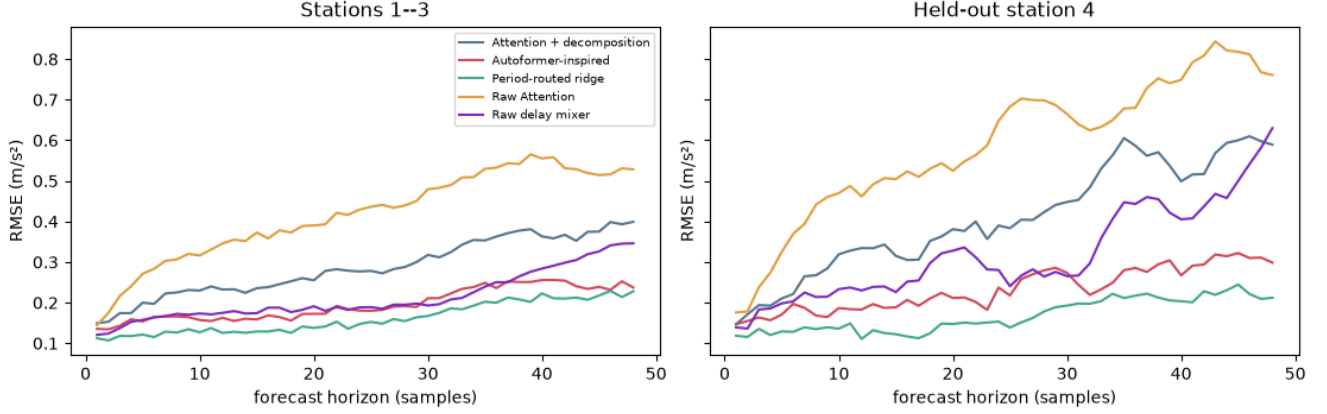


Figure 7: Output 4.1: largest combined-model gain (Station 3/B) and loss (Station 2/A); aggregate superiority is not universal.

Table 5: Output 4.1 selected-window RMSE, retained separately from aggregate comparisons.

Window	Attention	Raw delay	Decomp. Attention	Combined
Largest gain, Station 3/B	1.325	.479	.242	.139
Largest loss, Station 2/A	.206	.139	.285	.358

**Figure 8:** Output 4.2: validation error profiles over all 48 steps; curves need not increase monotonically.**Table 6:** Output 4.2 validation RMSE endpoints; complete curves are retained in the figure.

Model	Established		Held-out	
	Step 1	Step 48	Step 1	Step 48
Period-routed ridge	.112	.228	.119	.212
Raw Attention	.144	.528	.176	.761
Raw delay mixer	.121	.346	.139	.630
Attention + decomposition	.149	.399	.147	.589
Autoformer-inspired	.135	.237	.147	.299

Table 7: Output 4.3: final evaluation after selecting period-routed ridge for both populations.

Population	Validation MSE	Validation RMSE	Test MSE	Test RMSE
Established	.028	.166	.026	.161
Held-out	.030	.173	.033	.183

4(b–c), interpretation (under 200 words). Both switches help. Established MSE reductions from decomposition are .099 with Attention and .009 with delay mixing; delay mixing reduces it by .137 on raw and .047 on decomposed inputs. Held-out reductions are respectively .189/.059 and .258/.128. Delay mixing alone therefore has the larger aggregate gain, but these scores do not isolate drift-prominent contexts. Combined gains (.146 established, .317 held-out) are below summed gains (.236/.447), suggesting overlapping benefits without proving cause.

Autoformer-inspired is best overall among neural models, not universally: established Product A favors raw delay mixing (.211 versus .226), as does the contrasting loss window (.139 versus .358). Errors generally rise with horizon, especially for raw models on station 4; combined and routed forecasts remain comparatively stable.

Choose period-routed ridge for both populations: lowest validation error, 59,904 parameters, and lowest fitting time among competitive models. Validation-to-test RMSE is .166→.161 established and .173→.183 held-out. Test results did not determine selection. This single synthetic seed supports the benchmark conclusion, not universal real-world superiority.

2 Task 2 – Autoformer leaderboard challenge

2.1 Data, exploration, and hypotheses

The anonymized history has 43,656 observations; ten optional covariates cover 43,824 positions. Exactly 168 forecasts are required for `time_idx=43657-43824`. Target units and dates are unknown. Saved environment: PyTorch 2.11.0+cu130, CUDA, one CPU thread; seeds 0/1/2 are used for every main screen/ablation. Mean, standard deviation, median, quartiles, minimum, and maximum of the target are 98.17, 90.83, 73, 30/136, 0, and 994 (skewness 1.83).

Design hypotheses. Heavy tails motivate identity/square-root/log transforms, compared on raw-unit metrics rather than assumed optimal. Short-lived autocorrelation motivates testing known-future external information in the decoder. A small Autoformer may match or beat a wider one when parameter/epoch penalties matter. These are the notebook’s stated rationales; saved outputs alone do not establish their preregistration.

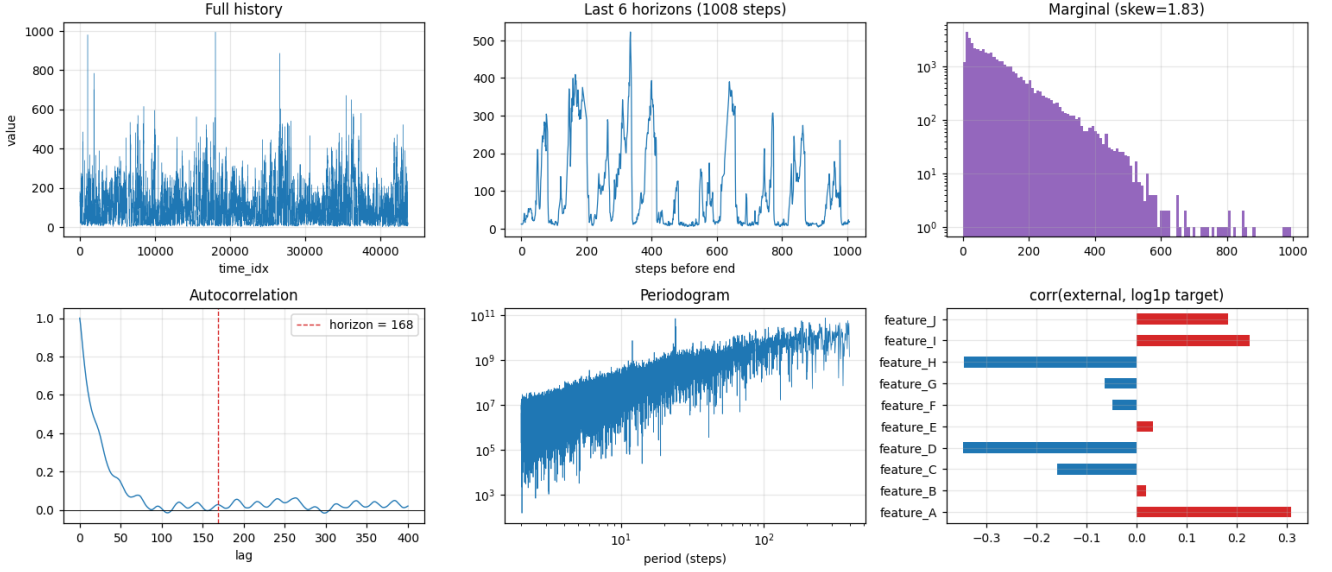


Figure 9: Task 2 exploration: complete/recent history, marginal distribution, ACF, periodogram, and correlations with the log-transformed target.

The recovered cycle is 24 steps; ACF at lags 1/24/168 is .966/.401/.028 and first falls below .1 at lag 55. This indicates weak long-horizon linear persistence, not proof that endogenous forecasting is impossible. Correlations suggest external signal (notably A, D, H, I, J) but are not causal evidence. Fourier features use 24, 168, and 365.25×24 -step cycles, each with two harmonics. A 24-step pattern *does not identify the physical sampling interval*; “day/week/year” are assumed feature labels, not recovered calendar dates.

Table 8: Task 2 eight strongest reported periodogram components (full observed history).

Period	Power	Period	Power	Period	Power	Period	Power
10914	.012139	2910.40	.010518	8731.20	.009689	269.48	.009309
24	.008909	7276	.008623	386.34	.007207	183.43	.007068

2.2 Model, preprocessing, and chronological evaluation

The from-scratch Autoformer follows Wu et al. and the Task 1 decomposition/delay formulation: two encoder layers, one decoder layer, four heads, progressive replicate-padded moving averages (width 25), and FFT Auto-Correlation with top- k delayed-value aggregation. Decoder self/cross mixing accumulates removed trends; it initializes future seasonal values to zero and trend to the context mean. Fixed sinusoidal positions and a linear covariate stamp supplement circular value convolutions. Context/warm-start/horizon lengths are 168/168/168. Delays are per-example, lag zero is excluded, and aggregation is implemented as FFT circular convolution rather than pointwise attention or a materialized delay gather. Unlike Task 1, Task 2’s projected query/key FFT scores are not explicitly mean-centered. Saved verification checks pass for direct versus FFT scores, positive-shift aggregation, and reconstruction/constant-signal decomposition.

Final width is 8, feed-forward width 16, dropout 0, AdamW learning rate 10^{-3} , weight decay 10^{-4} , batch 64, clipping 1, and 5% warm-up followed by cosine decay. Training uses stride 1. Square-root targets are standardized globally on the early fitting slice; a calibration-only affine inverse correction is used. Per-window normalization is avoided so absolute level remains available.

The full external configuration has 48 features: ten levels (D/E/F log-transformed), six differences, twelve trailing 6/24-step means, eight binary-regime $\times$ 24-step phase interactions, and twelve Fourier features. Past stamps enter the encoder and known-future stamps enter the decoder. The supplied future external measurements are permitted by the assignment; no future target is provided.

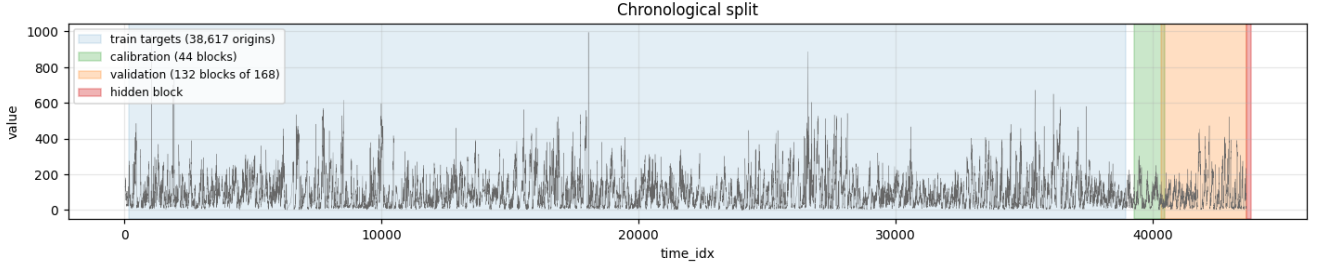


Figure 10: Saved chronological split. Training is separated from held-out contexts; the calibration and validation target bands partially overlap (see caveat).

Table 9: Saved Task 2 target spans, using zero-based half-open array positions.

Role	Origins	Stride	Union of target positions
Training	38,617	1	[168, 38952)
Affine calibration	44	24	[39288, 40488)
Reported validation	132	24	[40344, 43656)
Hidden forecast	1	–	[43656, 43824)

Evaluation caveats. Statistics are fitted before index 39,120; train targets end 168 steps before the first held-out context. However, calibration targets extend to 40,488 while validation targets begin at 40,344: **144 target samples overlap**. The saved assertion checks origin ordering, not disjoint target intervals. The reported numbers are retained as obtained, not labeled leakage-free independent validation. The 132 rolling windows also overlap each other; seed standard deviations measure initialization variation, not uncertainty across independent test blocks. Full-history EDA and repeated screens use the same observed holdout, so these results are model-selection estimates, not unseen-test scores.

2.3 Baselines, sanity check, and design screens

Table 10: Task 2 baseline and smoke-run validation. sMAPE is percent; MAE/RMSE are anonymized raw units. The smoke run is diagnostic, not multi-seed comparative evidence.

Model	MAE	RMSE	sMAPE
Persistence	93.091	133.895	90.309
Seasonal naive (168-step lag in code)	94.273	133.381	96.402
24-step cycle climatology	72.590	94.919	82.123
One-epoch smoke run, seed 0	60.275	85.259	71.573

The smoke fit uses width 32, 32,705 parameters, stride 12, 6.2 seconds, and normalized training loss 1.0336. The saved “seasonal naive” code repeats the prior 168-step block, not a 24-step cycle; the report uses that implemented baseline rather than its inconsistent prose description.

Table 11: Target-space screen: three seeds, four epochs, width 32, 32,705 parameters.

Transform	Mean RMSE $\pm$ SD	Mean MAE	Mean sMAPE	Ensemble RMSE	Fit s
Identity	68.780 $\pm$ 1.079	46.804	59.778	66.717	106.741
Square root	64.701 $\pm$ 1.691	43.115	55.173	62.159	106.776
Log1p	65.184 $\pm$ 5.987	44.426	58.331	60.716	107.255

Square root wins the *mean individual-seed* RMSE criterion; log1p has lower ensemble RMSE but substantially larger seed spread. The small mean difference is not strong evidence of universal transform superiority. RMSE emphasizes spikes, MAE typical errors, and sMAPE relative errors at low levels; a mean-shaped forecast can improve RMSE without equally improving sMAPE.

Table 12: Width screen: square-root targets, three seeds, four epochs. Assumed score includes $\alpha = 2 \times 10^{-5}$ and $\beta = .05$, not undisclosed official penalties.

Width	Parameters	RMSE $\pm$ SD	MAE	sMAPE	Ens. RMSE	Fit s	Assumed score
8	2,801	63.347 $\pm$ 1.014	42.622	54.147	61.151	105.625	63.603
16	9,185	66.079 $\pm$ 1.108	43.772	56.069	64.384	105.426	66.463
32	32,705	64.701 $\pm$ 1.691	43.115	55.173	62.159	105.920	65.556

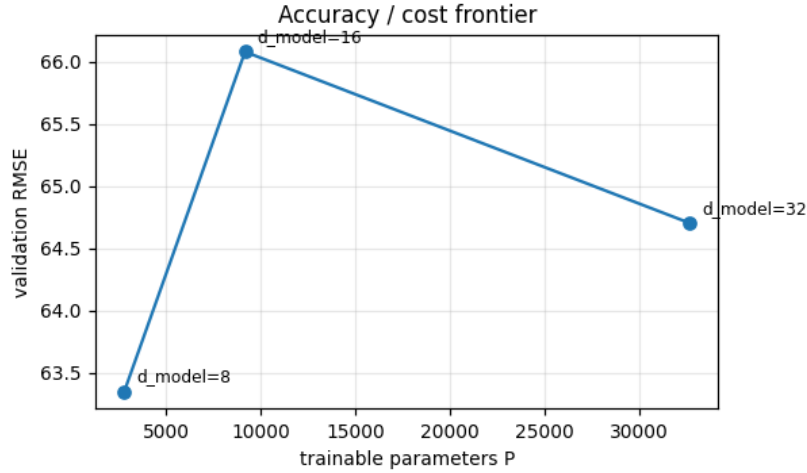


Figure 11: Width/cost frontier. Width 8 has lower mean error and fewer parameters than both larger models in the saved screen.

The width-16 to width-32 step buys .0000586 RMSE per parameter, but both are dominated by width 8. Printed RMSE per thousand parameters is 22.616/7.194/1.978 for widths 8/16/32; this ratio alone does not define efficiency, whereas lower error and lower total parameters justify width 8. True leaderboard penalties are unknown; assumed scores below are local decision proxies only.

2.4 Optional-external ablation across seeds

Table 13: Four-epoch ablation with square-root targets and width 8; seeds 0/1/2, identical origins. “Endogenous” retains Fourier features but uses no optional external values.

Features	Parameters	Mean RMSE $\pm$ SD	MAE	sMAPE	Ensemble RMSE
Endogenous only	2,225	99.810 $\pm$ 1.213	70.601	81.180	98.841
External levels	2,385	68.720 $\pm$ 1.320	47.832	59.393	67.152
External + differences	2,481	66.462 $\pm$ 2.943	46.604	59.294	64.595
External + differences + context	2,801	63.347 $\pm$ 1.014	42.622	54.147	61.151
Cycle climatology baseline	0	94.919	72.590	82.123	94.919

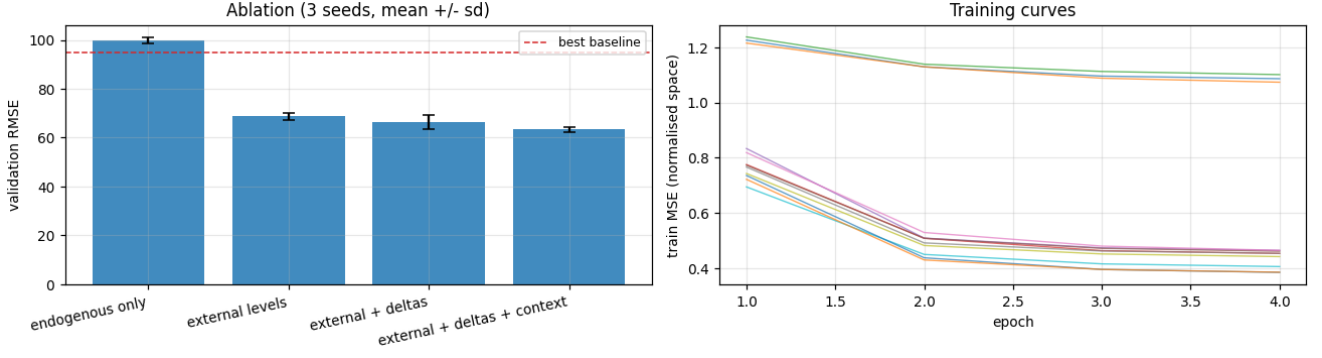


Figure 12: External-feature ablation and all three seeds’ training curves. Loss curves are in normalized transformed space, not raw-unit RMSE.

Full external context lowers mean RMSE by 36.463 (36.5%) versus endogenous-only and 31.572 (33.3%) versus climatology. The printed effect/pooled-seed-SD ratio is 23.1, a descriptive ratio, not a significance test. Levels provide the main gain; differences and engineered context further improve means, but their small incremental differences relative to seed spread warrant caution. Known-future information can sustain long-horizon forecasts where local target persistence has weakened. This is a collective-feature ablation, not causal attribution to individual variables.

2.5 Epoch budget, ensemble cost, and selected validation

Table 14: Epoch sweep for the selected external configuration (2,801 parameters per member).

Epochs	Mean RMSE $\pm$ SD	Ensemble RMSE	Mean MAE	Mean sMAPE	Mean fit s
2	65.916 $\pm$.935	64.166	44.801	57.426	52.924
4	63.347 $\pm$ 1.014	61.151	42.622	54.147	106.094
8	62.873 $\pm$.815	60.173	43.160	56.036	211.505

Table 15: All measured cost options under $S = \text{RMSE} + 2 \cdot 10^{-5} P + .05 E$. Single-member RMSE is the mean across seeds, not a separately selected seed.

Epochs/member	Members	Validation RMSE	Total P	Total E	Assumed S
8	3	60.173	8,403	24	61.541
4	3	61.151	8,403	12	61.919
8	1	62.873	2,801	8	63.329
4	1	63.347	2,801	4	63.603
2	3	64.166	8,403	6	64.634
2	1	65.916	2,801	2	66.072

The selected option is three members $\times$ eight epochs. Their ensemble gains exceed seed SD; 8 epochs improves RMSE but worsens mean MAE/sMAPE relative to 4, consistent with metric disagreement. For the same three-member size, 8 versus 4 epochs is preferable only if the per-epoch penalty is below approximately $(61.151 - 60.173)/12 = .0815$. Thus the choice depends on assumed, not official, cost coefficients. Selected inverse calibration: scale .9313, offset +17.506.

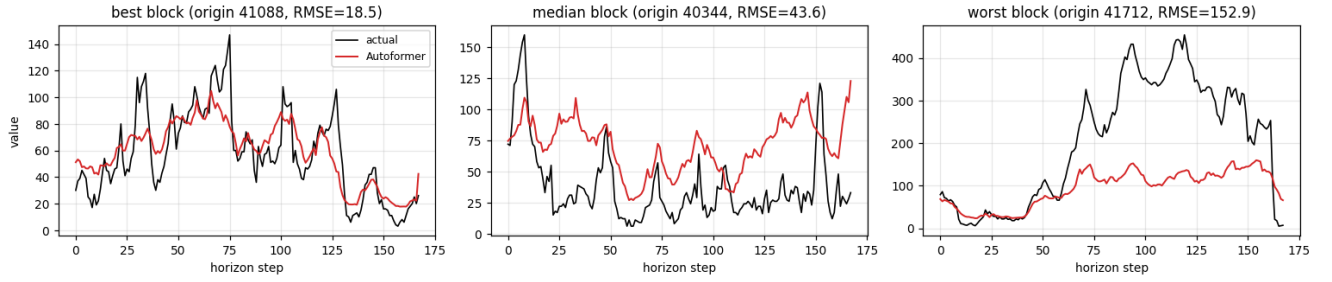


Figure 13: Selected validation ensemble: best/median/worst blocks at zero-based origins 41,088/40,344/41,712; RMSE 18.5/43.6/152.9.

Best-block shape is captured reasonably; the median block overpredicts low intervals, and the worst block strongly underpredicts sustained high values. Sharp, rare spikes remain difficult. The illustrated median-origin block also belongs to the calibration-overlap boundary region.

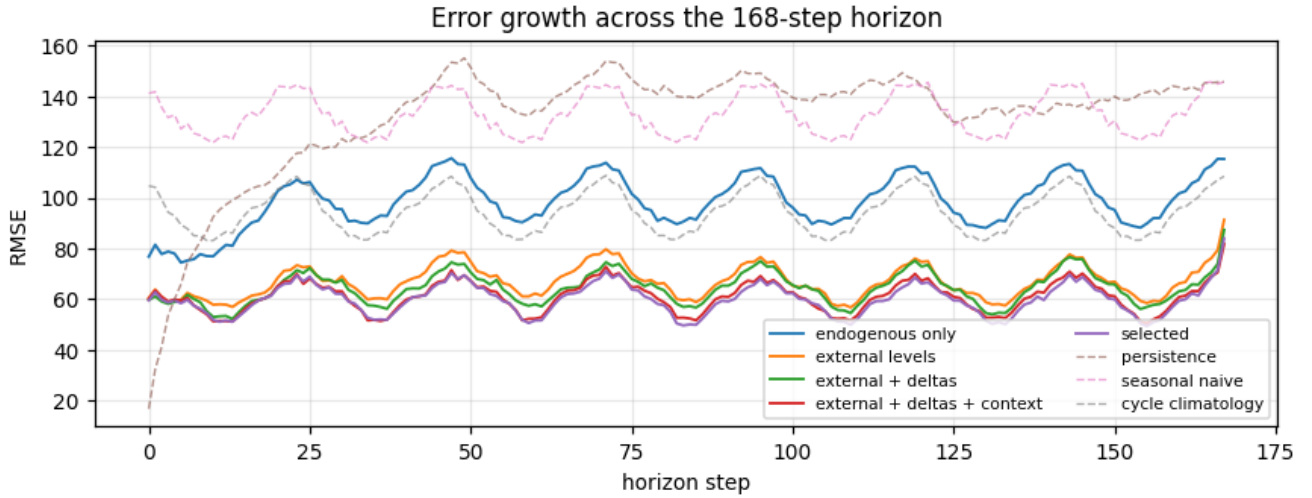


Figure 14: All ablation ensembles, selected 8-epoch ensemble, and baseline RMSE profiles across the 168-step horizon.

Endogenous-only error approaches/exceeds climatology after short lags; external configurations retain lower error across the horizon. Oscillatory error profiles reflect cycle-dependent difficulty and are not monotonically increasing. Ensemble RMSE 60.173 is about 36.6% below the best baseline; this is local rolling validation, not the hidden-block score.

2.6 Final fit, forecast, and artifact status

The final stage refits three new members on every usable observed-history target window (43,321 origins, stride 1), preserving the earlier normalizers and inverse-calibration pair. Seeds 0/1/2 each train eight epochs. Counted production parameters are $3 \times 2801 = \mathbf{8403}$ and epochs $3 \times 8 = \mathbf{24}$; exploratory screens are separate. The saved logs report final normalized training losses .3513/.3341/.3344 and fitting durations 235/233/233 seconds (about 701 seconds in total).

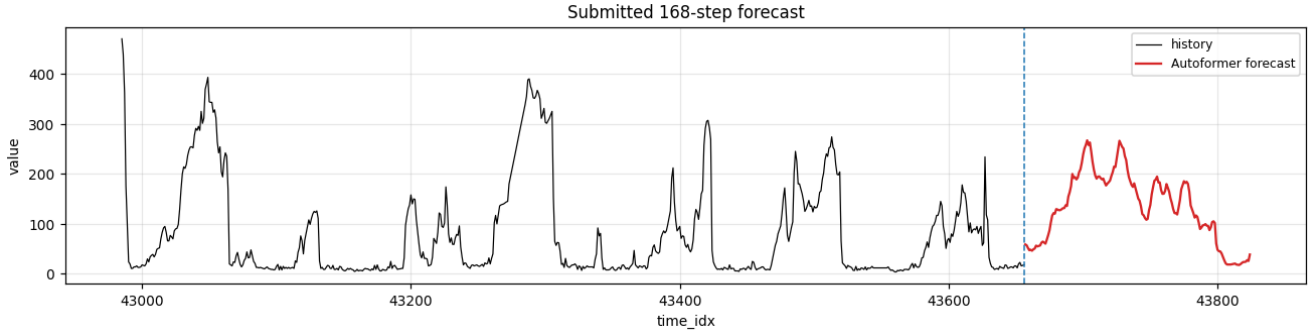


Figure 15: Saved final forecast over indices 43,657–43,824 beside observed history. The title says “Submitted” in the notebook; no public submission response is recorded.

Table 16: Saved final forecast and metadata; no hidden target is available.

Field	Recorded result
Horizon and ordering	168 values; 43657 first, 43824 last
Forecast range and mean	min 17.83; mean 132.71; max 267.06
Configuration	square root, width 8, full externals, 3 seeds, 8 epochs/member
Total production resources	$P = 8,403$; $E = 24$
Local individual-seed metrics	RMSE $62.873 \pm .815$; MAE 43.160; sMAPE 56.036%
Local ensemble metric	RMSE 60.173 (ensemble MAE/sMAPE not printed)
Logged export names	<code>predictions.txt</code> , <code>student_test_filled.csv</code> , <code>submission_meta.json</code>
Actual local artifact folder	Empty in the supplied q2-colab copy
Leaderboard/hidden-test outcome	Not recorded; official score unknown

The saved assertions report finite, nonnegative forecasts of length 168. The payload log displays only its initial values, beginning 58.9451, 57.2044, 51.7855, 47.2768; it cannot independently recover or verify the complete submission files. Some final cells retain outputs despite null execution counts. Validation averages individually inverse-transformed/calibrated predictions, whereas final production averages transformed predictions before the nonlinear inverse and uses the mean calibration pair. Those aggregation rules differ; the recorded 60.173 is not a direct evaluation of the final production averaging rule. No new metrics or missing forecasts are invented.

3 Combined conclusions and limitations

Task 1 rewards a matched period-conditioned linear bias over extra flexibility; its two neural upgrades still improve aggregate accuracy. Task 2 instead shows strong gains from permitted known-future covariates and an economical Autoformer, while external spikes remain challenging. The cost trade-off must count every ensemble member and training epoch.

These conclusions concern the saved runs. Task 1 is single-seed synthetic validation plus final test; Task 2 is three-seed model-selection validation with overlapping targets, assumed cost weights, and no public leaderboard result. Spectral patterns do not reveal undisclosed physical units/calendar dates. Both the split caveat and artifact status should be resolved before treating the Task 2 estimates as independently validated submission evidence.

A Per-seed results retained from Task 2

The compact table preserves every distinct screen/ablation/budget run’s printed raw-unit metrics. Each triplet is **RMSE / MAE / sMAPE(%)**. The full-external width-8 four-epoch run repeats in the width screen, ablation, and epoch sweep with the same printed metrics; it is listed once. Target screens use width 32/full externals; other feature/budget rows use square-root targets, width 8. Values here retain the precision printed per seed, so averaging them can differ slightly from full-precision summary tables.

Table 17: Individual seeds underlying all main Task 2 comparisons.

Configuration	Seed 0	Seed 1	Seed 2
Identity / width 32 / 4 epochs	68.554 / 47.042 / 59.84	69.954 / 46.712 / 59.82	67.832 / 46.659 / 59.67
Square root / width 32 / 4 epochs	62.906 / 43.079 / 56.46	66.264 / 43.549 / 55.76	64.934 / 42.716 / 53.30
Log1p / width 32 / 4 epochs	62.226 / 43.652 / 57.77	61.251 / 42.104 / 55.49	72.074 / 47.523 / 61.73
Square root / width 16 / 4 epochs	67.053 / 42.991 / 53.25	64.874 / 43.273 / 56.20	66.309 / 45.052 / 58.75
Full externals / width 8 / 4 epochs	62.521 / 43.081 / 55.93	63.042 / 41.786 / 52.85	64.479 / 43.001 / 53.67
Endogenous only / 4 epochs	100.678 / 70.499 / 81.76	98.425 / 69.790 / 80.14	100.328 / 71.513 / 81.63
External levels / 4 epochs	68.016 / 45.896 / 57.75	67.901 / 48.415 / 60.87	70.243 / 49.185 / 59.56
External + deltas / 4 epochs	69.857 / 49.743 / 63.29	64.640 / 45.188 / 56.54	64.889 / 44.881 / 58.05
Full externals / 2 epochs	66.520 / 45.611 / 58.08	64.840 / 43.558 / 55.57	66.390 / 45.232 / 58.64
Full externals / 8 epochs	63.514 / 43.186 / 56.32	63.149 / 42.963 / 56.97	61.956 / 43.332 / 54.82

B Evidence and attribution

Source evidence is the Task 1 notebook/report and exported tables, and the Task 2 notebook’s saved source, text/table outputs, and seven embedded plots. All 15 report figures use the original saved image data; no figure or experimental result was regenerated by model execution. Task 2’s local figure folder contained versions not byte-identical to four notebook images; the embedded notebook images were used consistently as the authoritative snapshot.

The Autoformer implementation cites Wu et al. (2021) and the student’s Task 1 formulation; its documented per-example delay selection, exclusion of zero delay, FFT aggregation, and fixed positional encoding are departures from the original reference implementation.

References

- [1] Wu et al. (2021). *Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting*, Sections 3.1–3.2.
- [2] Vaswani et al. (2017). *Attention Is All You Need*.
- [3] Zeng et al. (2023). *Are Transformers Effective for Time Series Forecasting?*
- [4] Hyndman and Athanasopoulos. *Forecasting: Principles and Practice*, Section 5.10.
- [5] AI651 Assignment 1 (Fall 2026), Tasks 1–2; supplied Task 1 harness and saved student notebooks.